Proposal & Implementation Plan

Feature Specification: Field Visibility per Order Type

1. Executive Summary

Currently, the TMS booking, road assignment, and trip screens display all available inputs (~215 fields on booking, ~145 on assignment, ~74 on trip) regardless of the selected order type. This adds visual clutter and slows down data entry for planners registering straightforward shipments.

The **Field Visibility per Order Type** feature introduces dynamic, order-type-driven UI screens. Administrators can configure field visibility (Visible, Hidden, Mandatory) directly on each Order Type. Planners will experience clean, relevant screens tailored strictly to the specific type of shipment being processed.

2. Key Objectives & Benefits

- **Simplified Planner UI:** Planners see only the cards and fields relevant to the selected Order Type, reducing cognitive load and entry errors.
- **No New Concepts for Admins:** Field rules are configured directly on the Order Type form view under a dedicated *Fields* tab—eliminating the need for separate screen profile objects.
- **Enhanced Data Integrity:** Server-side mandatory field validation ensures required fields are filled before an order can be confirmed.
- **Backward Compatibility & Data Safety:** Existing field values are never lost if a field is later hidden; hidden fields with existing data remain visible in read-only mode with a dedicated indicator badge.

3. Detailed Functional Capabilities

A. Order Type Administration ('Fields' Tab)

- **Dedicated Notebook Tab:** Added to the transport.order.type form view, containing two side-by-side management columns: *Booking Screen Configuration* and *Road Assignment Screen Configuration*.
- **Granular Visibility & Mandatory Controls:** Includes Master Card Switches to toggle entire sections, Field Visibility Toggles (Visible/Hidden), and Mandatory Checkboxes (Required to Confirm).
- **Productivity Tools:** *Copy Settings* action to duplicate field rules from an existing order type in one click, and *Reset Defaults* to revert all fields to visible.

B. Dynamic Planner Experience

- **Instant Real-Time Filtering:** Changing the Order Type on the booking screen updates visible cards and fields dynamically without requiring a page reload.
- **Automated Card Hiding:** When all fields within a card section are hidden, the entire card container is hidden automatically to optimize layout space.
- **Read-Only Preservation:** If an existing order contains data in a field that has since been marked as hidden for that order type, the field is rendered read-only with a *'Hidden by order type'* indicator badge.
- **Order Confirmation Check:** Draft orders can be saved freely, but clicking **Confirm Order** triggers server validation for mandatory fields specified by the order type.

4. Core System Rules & Hierarchy

- **Core Fields Protection:** Essential fields (e.g., Customer, Origin City, Destination City, Pickup Date, Carrier) are defined as Core fields (core=True) and can never be hidden by administrators.

- **Layered Visibility Hierarchy:** 1. *Core Definitions* (Always visible) | 2. *Market Conditions* (e.g., US Route, Mexico RFC, DDT compliance) | 3. *Order Type Rules* (Configured per Order Type) | 4. *User Local Settings* (Personal gear menu preferences).
- **Trip Screen Configuration:** Trip field visibility is managed per Trip Type at the company configuration level (res.company).

5. Implementation Roadmap

Phase	Description & Deliverables	Scope
Phase 1	**Data Model & Catalogue:** • Create central field catalogue registry (field_catalogue.py). • Create transport.order.type.field.rule model. • Migration script for existing mandatory fields.	Backend Models
Phase 2	**Admin OWL Editor:** • Build OWL field_rules_editor component. • Update Order Type form views with notebook tabs (General, Reporting, Planning Criteria, Fields).	Admin UI
Phase 3	**Screen Resolver Integration:** • Develop client-side useFieldVisibility hook. • Wire dynamic visibility into Booking, Road Assignment, and Trip OWL components. • Real-time change handlers and card auto-hiding.	Planner UI
Phase 4	**Validation & Tour Integration:** • Server-side confirm validation for mandatory fields. • Update guided tour step filtering so hidden fields are automatically bypassed.	Logic & Quality
Phase 5	**Automated Testing & i18n:** • Python unit test suite for rules, copying, and migration. • HttpCase browser tour test suite for full UI validation. • Localization strings (sv_SE.po and .pot).	QA & Launch

6. Acceptance & Approval

Upon approval of this proposal and roadmap, development will commence according to Phase 1. Complete delivery will include unit test coverage, HttpCase tour verification, and a live demonstration on the staging environment.